Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/17/2019
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for Fundamental Research on an International Standard of Fire Safety in Space
Start Date:	03/06/2014	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: New end date is 12/31/2018 per NSSC information (Ed., 10/16/18) NOTE: New end date is 9/05/2018 per NSSC information (Ed., 4/12/18)		
Key Personnel Changes/Previous PI:	Prof. Carlos Fernandez-Pello is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA Principal Investigator is Prof. Osamu Fujita, Hokkaido University.		
COI Name (Institution):	Carey, Van Ph.D. (University of California Berkeley)		
Grant/Contract No.:	NNX14AF01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012. The objective of the proposed research program is to continue the experimental study of the flammability of wire materials in space exploration atmospheres and associated computational/theoretical tools to aid interpretation of test results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in spacecraft allows us to accurately elucidate the effect of the environment parameters on the ignition and flame spread over combustible material, and through them their potential fire hazard. Particularly important is the determination of the Limiting Oxygen Concentration (LOC) on flame extinction under spacecraft environments. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Another important aspect of the research is the effect of melting and dripping of plastic insulation in normal gravity in comparison with microgravity. The results are relevant because dripping will not occur in microgravity and consequently could impact their burning and methodology to screen. The investigation and results have also benefits for terrestrial fire safety by providing further information about the flammability of materials under a variety of environments.
Task Progress:	During this reporting period, melting and dripping behavior during the opposed flame spread over horizontal and vertical polyethylene (PE) insulated wires were investigated. Two different wire dimensions – core/wire diameter of 3.5/8.0 and 5.5/9.0 mm – and three different insulations – low-density polyethylene (LDPE), high-density polyethylene (HDPE), and black LDPE (B-LDPE) – were tested. In order to identify the effect of the core, wires with solid copper core (Cu), hollow stainless steel core (SS), and without core were tested. Experimental results showed that a high-conductance copper core acted as a heat source downstream to increase the flame spread rate. However, in the upstream burning region, the copper core acted as a heat sink, cooling and reducing the mobility of the melting insulation. As a result, the copper core extended the insulation’s residence time in the flame to facilitate burning and reduce dripping. Moreover, in the vertical downward flame spread, the insulation drips flowed downward to preheat the wire, and this heating dominated over the heating by the core. As a result, higher dripping fraction for wires with lower thermal conductance resulted in faster flame spread. This is the first time that the simultaneous dual effect of heat source and sink for wire core was observed, and the influence of dripping on flame spread over wire was discovered. NOTE that this project is continued by "Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space"; grant number 80NSSC19K0331 with the same Principal Investigator, Prof. Carlos Fernandez-Pello. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Articles in Peer-reviewed Journals	Kobayashi Y, Konno Y, Huang X, Nakaya S, Tsue M, Hashimoto N, Fujita O, Fernandez-Pello C. "Laser piloted ignition of electrical wire in microgravity." Proceedings of the Combustion Institute. 2019;37(3):4211–9. https://doi.org/10.1016/j.proci.2018.06.089 , Jan-2019
Articles in Peer-reviewed Journals	Lu Y, Huang X, Hu L, Fernandez-Pello C. "The interaction between fuel inclination and horizontal wind: Experimental study using thin wire." Proceedings of the Combustion Institute. 2019;37(3):3809-16. https://doi.org/10.1016/j.proci.2018.05.131 , Jan-2019
Articles in Peer-reviewed Journals	Lu Y, Huang X, Hu L, Fernandez-Pello C. "Concurrent flame spread and blow-off over horizontal thin electrical wires." Fire Technology. 2019 Jan;55(1):193-209. https://doi.org/10.1007/s10694-018-0785-0 , Jan-2019